

Exam - Time series forecasting

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	A	B	C
1	Timestamp	Power (kW)	Temp (C°)
2	1/1/2010 1:15	165,1	10,6
3	1/1/2010 1:30	151,6	10,6
4	1/1/2010 1:45	146,9	10,6
5	1/1/2010 2:00	153,7	10,6
6	1/1/2010 2:15	153,8	10,6
7	1/1/2010 2:30	159,0	10,6
8	1/1/2010 2:45	157,7	10,6
9	1/1/2010 3:00	163,2	10,6
10	1/1/2010 3:15	151,7	10,0
11	1/1/2010 3:30	148,7	10,0
12	1/1/2010 3:45	155,1	10,0
13	1/1/2010 4:00	161,5	10,0
14	1/1/2010 4:15	161,5	10,0

Figure 1: Elec-train.xlsx

The file **Elec-train.xlsx** contains electricity consumption (kW) and outdoor air temperature for one building. These quantities are measured every 15 minutes, from 1/1/2010 1:15 to 2/18/2010 23:45. In addition, outdoor air temperature are available for 2/19/2010. The goal is to forecast **electricity consumption (kW) for 2/19/2010**.

Part 1

The first part of the evaluation concerns the use and the comparison of all the forecasting methods studied during the course (exponential smoothing, SARIMA, Neural Net, LSTM, XGBoost, Random Forest, SVM. . .). So you have to test all these methods, to tune their hyperparameters and compare them properly, using cross-validation.

Then, once the best model has been selected, you should return its forecast, in one Excel file entitled **YourName.xlsx**, with **exactly** one column and 96 rows. Its quality, in comparison with the forecasts of the other students, will be take into account in the grading.

Part 2

In the second part of the evaluation, you should implement in R the WNN forecasting method (described in Section 3.1 of [1]) and then compare the WNN method with the forecast you did in part 1. You should return an R code containing the WNN function, with an example of its use on a simple time series (for instance co2).

Expected deliverable

In addition to your forecast in Excel and your R code for WNN, you should also return a short report (few pages in pdf), entitled **YourName.pdf**, explaining how you have proceeded to select the best forecasting model and presenting the comparison results.

Reference

[1] Talavera-Llames, R.L., Pérez-Chacón, R., Martínez-Ballesteros, M., Troncoso, A., Martínez-Álvarez, F. (2016). A Nearest Neighbours-Based Algorithm for Big Time Series Data Forecasting. In: Martínez-Álvarez, F., Troncoso, A., Quintián, H., Corchado, E. (eds) Hybrid Artificial Intelligent Systems. HAIS 2016. Lecture Notes in Computer Science, vol 9648. Springer, Cham.